

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds 4 or more correct row minima/col maxima	AO1.1a	M1	Row minima = (2, 0, -2) Col maxima = (2, 3, 5)
	Correctly finds all row minima/col maxima	AO1.1b	A1	Max(row minima) = 2 Min(col maxima) = 2
	Explains correctly that as the max(row minima) is equal to the min(col maxima) then a stable solution exists and the value of the game is 2	AO2.4	E1	As Max(row minima) = 2 = Min(col maxima), then a stable solution exists and the value of the game is 2
	Allow 'Maximin' and 'Minimax'			
	Marking Instructions	AO	Marks	Typical Solution
ALT (a)	Uses dominance to reduce the size of the pay-off matrix to 2×3 or 3×2	AO1.1a	M1	A_1 (or A_2) dominates A_3 , so remove A_3 [from pay-off matrix]
	Uses dominance to reduce the pay-off matrix to 1×1	AO1.1b	A1	S_1 [now] dominates S_2 and S_3 , so remove S_2 and S_3
	OR Correctly finds all remaining row minima/col maxima			A_1 [now] dominates A_2 , so remove A_2
	Explains that each player will play the same strategy each time which results in Alex gaining 2 each game OR Explains correctly that as the max(row minima) is equal to the min(col maxima) then a stable solution exists and the value of the game is 2 Allow 'Maximin' and 'Minimax'	AO2.4	E1	Alex will only ever play A_1 and Sam will only ever play S_1 , which results in Alex gaining 2 each game. Therefore, the value of the game is 2.
(b)	Identifies strategy for each player Condone stating ' A_1 ' and ' S_1 '	AO1.1b	B1	Alex strategy A_1 Sam strategy S_1
				Total 4 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies correctly all three row minima or all three (two) column maxima	AO1.1a	M1	Row minima: -1, -2, (-1) Column maxima: 4, 0, (1)
	Determines correctly max(row min) and min(col max)	AO1.1b	A1	max (row min) = -1 min (col max) = 0
	Makes correct statement for the play-safe strategies of each player	AO3.2a	A1 CAO	Victoria plays R (or P); Albert plays Y
(b)	Identifies strategy and provides justification with reference to dominance	AO2.4	E1	Albert should never play Z because strategy Y dominates strategy Z

(c) (i)	States correctly that strategy R dominates strategy P (PI)	AO1.1a	B1	Victoria should never play P because strategy R dominates strategy P
	Introduces and defines a probability variable	AO3.3	M1	Let Victoria play strategy Q with probability p and strategy R with probability $1 - p$.
	Finds correctly both expected gain expressions for Victoria	AO1.1b	A1	If Albert plays: X : expected gain for Victoria $= -2p + 4(1 - p) = 4 - 6p$
	Clearly states that, as the game has no stable solution, the optimal solution occurs when the two expected gain expressions are set equal to each other, and then solves correctly 'their' equation constructed using 'their' expected gains for Victoria	AO2.2a	A1F	Y : expected gain for Victoria $= -(1 - p) = p - 1$ No stable solution, so optimal value of p occurs when: $4 - 6p = p - 1,$ $p = 5/7$
	Interprets correctly the solution to the problem in the context, giving the optimal mixed strategy for Victoria	AO3.4	E1	Victoria should play strategy Q with a probability of $5/7$ and R with a probability of $2/7$
(ii)	Substitutes 'their' value of p into one of 'their' expected gain expressions, resulting in a	AO1.1b	B1F	$4 - 6 \times (5/7) = -2 / 7$ or $(5/7) - 1 = -2/7$

	value of the game			
(iii)	Recognises correctly that Victoria can improve on the value of the game if Albert does not play an optimal mixed strategy	AO3.5b	E1	The expected pay-off for Victoria will only be $-2/7$ if Albert plays an optimal mixed strategy between X and Y
Total 11 marks				

Q3.

(a) Min

$$\begin{pmatrix} 4 & -1 & 2 & 3 \\ 4 & 6 & 3 & 7 \\ 1 & 3 & -2 & 4 \end{pmatrix} \begin{matrix} -1 \\ 3 \\ -2 \end{matrix}$$

Max 4 6 3 7

Maximin (row) = 3
Either correct, including correct values

Minimax (col) = 3
Both correct, written as equations PI by next line

M1

A1
CSO

As Maximin (row) = Minimax (col)
 There is a stable solution
Must have equation and statement and scored first 2 marks

E1

(Play safe) (H) B
 (Play safe) (W) F }
Both correct

B1
4

(b) Saddle point (B, F)

B1
1

[5]

Q4.

(a) (i)

Row min $-6, \quad -3, \quad -5, \quad -4$

Max (row min) = -3

Col max $5, \quad 4, \quad -3$

Min (col max) = -3

*attempt to find maximin and minimax
condone one slip in values*

M1

**all rows min and col max values correct
and max (row min) = -3 identified
and min (col max) = -3 identified**

A1

max (row min) = min (col max) = -3

$\Rightarrow$ game has a stable solution

**full statement involving maximin and minimax and both
values = -3**

E1

3

(ii) Adam plays **A₂** & Bill plays **B₃**

B1

1

(iii) Value of game for Bill is **+3**

B1

1

*Examiners must use the correct symbol for marks carried
forward, ie ringed totals with arrows through them.*

(b) (i) (Never play) **C₂**

correct strategy stated

C₂ dominated **by** C₁ ($-3 > -4$ and $2 > 1$)

and correct reason

condone $3 < 4$ and $-2 < -1$

B1

1

(ii) C₁ : $3p - 2(1-p)$

either correct unsimplified

M1

$$C_3 : -3p + 5(1-p)$$

both correct unsimplified
 $\{5p - 2, 5 - 8p\}$

A1

2

(iii) $3p - 2(1-p) = -3p + 5(1-p)$
equating their 2 expressions

M1

$$\Rightarrow p = \frac{7}{13}$$

A1

2

(iv) Value of game $= 5 \times \frac{7}{13} - 2$
 or $5 - 8 \times \frac{7}{13}$
 $= \frac{9}{13}$

B1

1

[11]

Q5.

- (a) (i) Row minima 2, -3, x

Check for answers written on table

B1

1

- (ii) Column maxima 3, 6, 4

Check for answers written on table

B1

$$\text{Max (row min)} = 2$$

$$\text{Or } 2 \neq 3$$

$$\text{Min (col max)} = 3$$

Condone Best (worst) = 2 etc

Worst (best) = 3

M1

Since $2 \neq 3 \rightarrow$ no stable solution



Both lines and statement must score
previous B1, B1

A1cso

3

(b)

$$\left. \begin{array}{l} x < 2, x + 3 < 6, 3 < 4 \\ \rightarrow R_1 \text{ dominates } R_3 \end{array} \right\} \text{Either of these}$$

hence Rhona should not play R_3

B1

1

- (c) (i) Let Rhona play R_1 with prob p
and R_2 with prob $1 - p$

When C plays

$$\begin{aligned} C_1 : \text{exp value} &= 2p + 3(1 - p) \\ &= 3 - p \end{aligned}$$

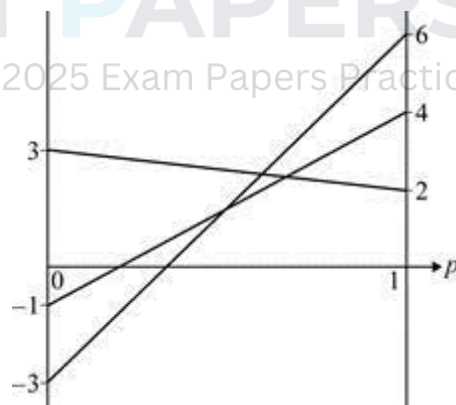
$$\begin{aligned} C_2 : 6p - 3(1 - p) \\ &= -3 + 9p \end{aligned}$$

$$\begin{aligned} C_3 : 4p - (1 - p) &= -1 + 5p \\ &\text{any two correct unsimplified} \end{aligned}$$

M1

all correct unsimplified

A1



drawing two of their expected values for $0 \leq p \leq 1$
both vertical axes using same scale
condone use of horizontal lines in paper

M1

all three correct lines
must see numbers on at least one vertical axis

A1

$$3 - p = -1 + 5p$$

choosing highest point of region

M1

$$\rightarrow p = \frac{2}{3}$$

A1

$\rightarrow$ Rhona plays R_1 $\frac{2}{3}$ of time
and R_2 $\frac{1}{3}$ of time
ft their p

E1✓

7

(ii) Value of game $= 3 - \frac{2}{3} = \frac{7}{3}$
or $-1 + \frac{10}{3} = \frac{7}{3}$

B1

1

[13]

Q6.

- (a) Row minima are $-4, -3, -7$
both row minima **and** column maxima attempted
(condone 2 errors)

©2025 Exam Papers Practice. All rights reserved.

M1

Column maxima are $-3, 6, 8$
all values correct

A1

$\max(\text{row min}) = \min(\text{col max}) = -3$
*condone arrows pointing to this element but
must **state** max (row min) and min (col max)
or equivalent*

E1

Play-safe Tom II and Jerry **A**

B1

4

- (b) (i) Let Rohan play R_1 with prob p
 $\Rightarrow$ plays R_2 with prob $1 - p$

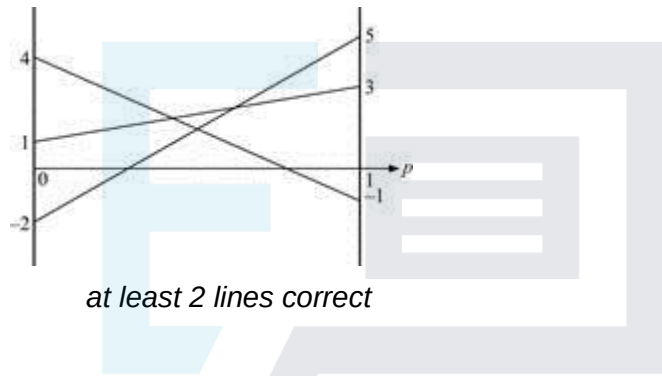
When Carla plays C_1 ,
 Rohan's expected gain = $3p + (1 - p)$
 $= 1 + 2p$

$C_2 : 5p + (-2)(1 - p) = 7p - 2$
at least 2 expected gains correct unsimplified

M1

$C_3 : -p + 4(1 - p) = 4 - 5p$
all 3 correct unsimplified

A1



M1

all lines correct for $0 \leq p \leq 1$ and values at 0 and 1 clear

A1

$$7p - 2 = 4 - 5p$$

$$12p = 6$$

choosing highest point or using correct equation

M1

$$\Rightarrow p = \frac{1}{2} \Rightarrow$$

Rohan plays R_1 50% of the time and
 R_2 50% of the time

A1cso

$$\text{Value of game} = 7 \times \frac{1}{2} - 2 = \frac{3}{2} \quad \text{AG}$$

$$\text{or } 4 - \frac{5}{2} = \frac{3}{2} \quad \text{must see working}$$

B1

7

- (ii) When Rohan plays R_1 , expected loss for

Carla is $3p + 5q + (-1)(1 - p - q)$

and when Rohan plays R_2 , expected loss
for Carla is $p + (-2)q + 4(1 - p - q)$
either expression correct unsimplified

M1

$$4p + 6q = \frac{3}{2} + 1$$

$$3p + 6q = 4 - \frac{3}{2}$$

correct simultaneous equations unsimplified

A1

$$\Rightarrow p = 0, q = \frac{5}{12}$$

condone 0.42 or better

A1

$\Rightarrow$ Carla never plays C_1 ,

plays C_2 with prob $\frac{5}{12}$

and plays C_3 with prob $\frac{7}{12}$

Must have all 3 correct probabilities

E1cso

©2025 Exam Papers Practice. All rights reserved.

4

[15]

Q7.

(a)

				Row min
				-2
				-3
				-5
Col max	4	0	-2	
	<i>Row minima and column maxima (all values)</i>			

B1

Max (row min) = -2

Min (col max) = -2

Both attempted or stated/indicated

M1

Since these are equal, there is a stable solution

*Must have both values = -2 plus
statement (withhold if max (min) and min
(max) not stated)*

A1

Ann plays A_1 and Bill plays B_3 for play-safe

E1

4

- (b) (i) Let Russ play R_1 with probability p
And R_2 with probability $1 - p$

C_1 : expected gain $-4p + 2(1 - p)$
 $(2 - 6p)$

C_2 : $7p - (1 - p) = 8p - 1$
2 correct unsimplified

M1

C_3 : $3p + (1 - p) = 1 - 4p$
All correct

A1



Plotting 3 expected gains for $0 \leq p \leq 1$

M1

Correct gains plotted accurately

A1

Solving $8p - 1 = 1 - 4p$

Choosing highest point of their region or correct

M1

$$\Rightarrow p = \frac{1}{6}$$

A1

$\Rightarrow$ Russ plays R_1 with probability $\frac{1}{6}$
and R_2 with prob $\frac{5}{6}$

E1

7

(ii) Value of game = $\frac{8}{6} - 1$
Or $1 - \frac{4}{6}$
 $= \frac{1}{3}$

B1

1

[12]

Q8.

(a)

Row minima
-7
-1
-3

Column max 6 8 -1
Max (row min) = -1

*Attempting Row Min & Col Max
or Maxmin and Minmax*

M1

©2025 Exam Papers Practice. All rights reserved.

Min (col max) = -1

*All values correct and shown with
correct words*

A1

Since these values are equal the game has
a stable solution

Must both be -1 and have statement

E1

Raj plays II, Cal plays Z

B1

4

(b) (i) $C_1 : 5p - 2(1 - p)$
 $7p - 2$

B1

$$C_2 : xp + 4(1 - p)$$

B1

2

(ii) Value of game = $\frac{8}{3}$
 $\Rightarrow 5p - 2(1 - p) = \frac{8}{3}$

$$\text{Their expected gain } (C_1) = \frac{8}{3}$$

M1

$$\Rightarrow p = \frac{2}{3}$$

A1

$$xp + 4(1 - p) = \frac{8}{3}$$

$$\text{Their } C_2 \text{ gain} = \frac{8}{3} \text{ (must involve } x\text{)}$$

M1

$$\Rightarrow \frac{2}{3}x + \frac{4}{3} = \frac{8}{3}$$

$$\Rightarrow x = 2$$

A1

4

©2025 Exam Papers Practice. All rights reserved.

[10]

Q9.

- (a) (For each outcome)
 Rowena's gain + Colin's gain = 0
One player's loss is other's gain

E1

1

- (b) (Column maxima 2, 5, 4)

$\Rightarrow \min(\text{colmax}) = 2$ (OE but strict)
Withhold E mark if any value incorrect; accept column minimax = 2

E1

⇒ Colin's play-safe strategy is C_1

B1

2

(c) R_3 is dominated by R_1

$$-5 < -4; \quad 4 < 5 \text{ and } 3 < 4$$

$E0$ if R_2 mentioned as well

E1

1

(d) Let Rowena play R_1 with prob p
and R_2 with prob $1 - p$

Expected gain when Colin plays

$$C_1: -4p + 2(1 - p) = 2 - 6p$$

$$C_2: 5p - 3(1 - p) = -3 + 8p$$

$$C_3: 4p - (1 - p) = -1 + 5p$$

attempt at least 2 with one correct

all 3 correct unsimplified

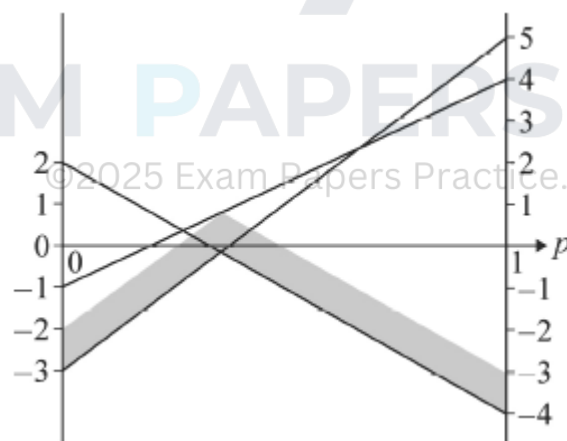
M1

A1

Plot expected gains against p for $0 \leq p \leq 1$

All 3 drawn ft their exp gains

M1



correct

A1

$$\Rightarrow 2 - 6p = -3 + 8p$$

Using "correct" equation

Choosing highest point of region

M1

$$\Rightarrow p = \frac{5}{14}$$

A1

Therefore Rowena plays R_1 with prob $\frac{5}{14}$
and R_2 with prob $\frac{9}{14}$
ft their p

E1ft

7

[11]

Q10.

(a) (i)

Row min

-4

-2

-1

Col max 5 -1 3

Attempt at row minimum and column maximum

M1

all figures correct

A1

$\min(\text{col max}) = \max(\text{row min})$
 $\Rightarrow$ stable solution

$\min(\text{col max}) = \max(\text{row min})$
 $\Rightarrow$ stable solution

E1

3

(ii) Ros plays III and Col plays Y

B1

value of game = -1

B1

2

(b) (i) Ros plays R_1 with probability p and R_2 with probability $1 - p$

Expected gains when Col plays:

$$C_1: 3p - 2(1 - p) = 5p - 2$$

$$C_2: 2p - (1 - p) = 3p - 1$$

$$C_3 : p + 2(1 - p) = 2 - p$$

attempt at least 2

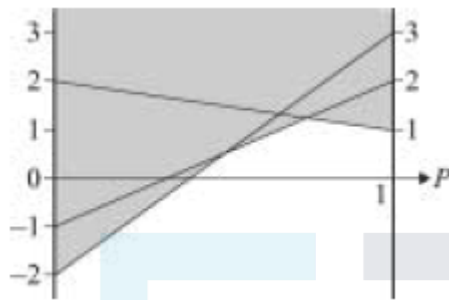
M1

correct unsimplified

A1

Plot expected gains against p for $0 \leq p \leq 1$

M1



correct (must see 0 or 1 on P axis, or implied by their numbers)
A0 if not possible to see highest point of region being correct

A1

Choose highest point of region below
 lines $\Rightarrow 3p - 1 = 2 - p$

must be this pair of lines or their highest point

M1

leading to $p = \frac{3}{4}$

©2025 Exam Papers Practice. All rights reserved.

A1

Therefore Ros plays R_1 with prob $\frac{3}{4}$
 and plays R_2 with prob $\frac{1}{4}$
ft their p from any lines

B1ft

7

(ii) Value of game $= 3 \times \frac{3}{4} - 1$
 or $\left(2 - \frac{3}{4}\right) = 1\frac{1}{4}$

B1

1

[13]

Q11.

- (a) (i) $\text{Min } R_1 (5, 2, -1) = -1$
 $\text{Min } R_2 (-3, -1, 5) = -3$
 $\text{Min } R_3 (4, 1, -2) = -2$

E1

Max min = -1
 $\Rightarrow$ Play safe strategy R_1

B1

2

- (ii) $\text{Max } C_1 = 5; \text{max } C_2 = 2; \text{max } C_3 = 5$
 $\text{Min } (5, 2, 5) = 2$

M1

$2 \neq 1 \Rightarrow$ no stable solution

A1

2

- (b) $R_3 (4, 1, -2) < R_1 (5, 2, -1)$

E1

1

- (c) (i) C_1 played, expected gain for Rose:

$$5p + -3(1 - p)$$

Any correct expected gain unsimplified

©2025 Exam Papers Practice. All rights reserved.

M1

$$= 8p - 3$$

One correct simplified

A1

$$C_2 \div p - (1 - p) = 3p - 1$$

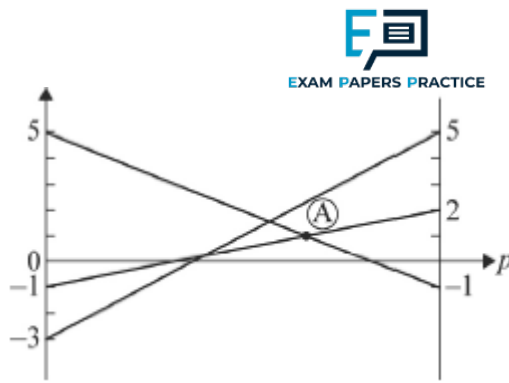
$$C_3 \div -p + 5(1 - p) = 5 - 6p$$

All correct simplified

A1

3

- (ii) Expected gain



Plotting at least 2 lines

M1

All correct with values at $p = 0$ and $p = 1$ indicated

A1

2

(iii) Choosing A – highest point in feasible region

$$\Rightarrow 3p - 1 = 5 - 6p$$

$$9p = 6$$

Solving this equation

M1

$$\Rightarrow p = \frac{2}{3}$$

CSO

A1

$\Rightarrow$ Rose plays R_1 $\frac{2}{3}$ of time
and R_2 $\frac{1}{3}$ of time

©2025 Exam Papers Practice. All rights reserved.

E1ft

3

(iv) Value of game $= 3 \times \frac{2}{3} - 1 = 1$
Or $5 - 4 = 1$

B1

1

[14]

Q12.

(a) $(-2, 2, 4) < (2, 4, 5)$

So S_1 dominated by S_2

E1

$$\begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix} > \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}$$

note > sign

So C_3 dominated by C_2

E1

2

(b) $C_1 \quad C_2$

$$2 \times 2 \text{ game now } \begin{matrix} S_2 & \begin{bmatrix} 2 & 4 \end{bmatrix} \\ S_3 & \begin{bmatrix} 5 & 1 \end{bmatrix} \end{matrix}$$

Minimum of rows $(2, 4) = 2$

correct method for either S or C

M1

Minimum of $(5, 1) = 1$

Choose maximum = 2

play safe for Sam is S_2

A1

Max of column 1 = $\max(2, 5) = 5$

Max of column 2 = $\max(4, 1) = 4$

Choose minimum = 4

play safe for computer is C_2

A1

©2025 Exam Papers Practice. All rights reserved.
 Since $2 \neq 4 \Rightarrow$ not stable solution

E1

4

(c) (i) Computer picks C_1

$$\text{Expected game} = 2p + 5(1 - p)$$

M1

$$= 5 - 3p$$

A1

Computer picks C_2

$$\text{Expected gain} = 4p + (1 - p)$$

$$= 1 + 3p$$

A1

3

(ii) Best mixed strategy

$$5 - 3p = 1 + 3p$$

M1

$$\Rightarrow p = \frac{2}{3}$$

M1 A1

2

(iii) Expected points gain

$$= 5 - 3 \times \left(\frac{2}{3}\right)$$

$$\text{Or } 1 + 3 \left(\frac{2}{3}\right)$$

$$= 3$$

B1

1

[12]

Q13.

(a)

Min rows

$$\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

Max = 2

M1

Max cols Min = 4

(7, 4, 6)

A1

$2 \neq 4$

E1

3

(b)

III > II

B1

$$\begin{bmatrix} 1 & 3 & 6 \\ 7 & 4 & 2 \end{bmatrix}$$

B1

2

- (c) Let Bev play with
 $p, q, 1 - p - q$

M1

$$p + 3q + 6(1 - p - q) = 3\frac{3}{5}$$

A1 without $\left(= 3\frac{3}{5}\right)$

A1

$$7p + 4q + 2(1 - p - q) = 3\frac{3}{5}$$

A1

$$-5p - 3q = -2\frac{2}{5}$$

$$5p + 2q = 1\frac{3}{5}$$

$$\Rightarrow q = \frac{4}{5}, p = 0$$

A1

Plays I prob 0

II prob $\frac{4}{5}$

III prob $\frac{1}{5}$

All three

©2025 Exam Papers Practice. All rights reserved.

B1

5

[10]

Examiner reports

Q1.

Nearly 90% of students scored at least one mark in part (a). Of these students, many scored 2 marks but failed to score the third mark due to a lack of depth in their explanation of why the value of the game was 2. Many did not refer to, or were not clear enough about, the $\max(\text{row minima})$ and $\min(\text{col maxima})$ both being equal to 2 and therefore being equal to each other, and many did not then relate this back to the value of the game.

Interesting correct solutions included the use of dominance to reduce the size of the pay-off matrix to 1×1 . Of the errors that students made, the most prevalent was determining the row maxima and column minima, which lead to an incorrect conclusion and often impacted the answer to (b).

Q3.

Although the majority of students had an idea of the requirements of this question, full marks were rarely achieved. It is essential that students give full and clear statements to provide evidence to the examiner; simply writing down seven numbers, circling two of the numbers, then stating ' $3 = 3$ ' will not score full marks. Students needed to show the seven numbers, but then they had to state; ' $\maximin(\text{row}) = 3$ ' and ' $\minimax(\text{col}) = 3$ ', followed by ' $\text{as } \maximin(\text{row}) = \minimax(\text{col}), \text{ there is a stable solution}$ '. The play-safe strategies were well-answered, but there were very few correct answers for the saddle point.

Q4.

- (a) (i) Explanations rarely scored full marks. In order to show that the game has a stable solution, it was expected that the maximum values in the columns would be indicated before finding the minimum value of these maxima. The maximum of the row minima also needed to be found. Some statement should then have been made indicating that these two values were equal and hence that the game has a stable solution. Quite a few students felt that all they needed to do was to draw arrows pointing to -3 when more detail was required. A few chose to solve the problem using an argument based on dominance, but the explanations were rarely adequate to earn more than a single method mark.
- (ii) It was encouraging to see many students correctly identify the play-safe strategy for each player.
- (iii) Only the better students stated the correct value of the game for **Bill**, despite the bold type; as you might expect weaker students gave the value as -3 , and others gave the value as 0, possibly confused by the term "zero-sum game".
- (b) *A large number of students were not prepared for this part of the question.*
- (i) A large number deleted the first column and scored only a couple of method marks for the rest of this question. Some who realised that the computer should not play C_2 spoiled their answers by writing such things as "each of the other strategies is better".
- (ii) The better students managed to use the values in the table correctly to find correct expressions for the expected gains for Roza for the two remaining strategies. A few ignored the dominance from part(b)(i) and attempted to solve

the problem graphically using three lines; some successfully, others not.

- (iii) Some poor algebra, usually when removing brackets, prevented some from finding the correct value of p , but most marks were lost for having the incorrect pair of simultaneous equations.
- (iv) Those who scored full marks in parts (b)(ii) and (iii) were usually able to obtain the correct value of the game for Roza.

Q5.

In part (a)(i) the entry “x” caused problems to several candidates. Credit was generously given to those who wrote the correct three minima at the ends of the rows in the pay-off matrix, when they were perhaps unaware of what was required in the request made in this part.

In part (a)(ii) explanations rarely scored full marks. In order to show that the game did not have a stable solution, it was expected that the maximum values in the columns would be indicated before finding the minimum value of these maxima. The maximum of the row minima from part (a)(i) also needed to be found. Some statement should then have been made indicating that these two values were not equal and hence that the game did not have a stable solution.

In part (b)(i) most candidates were able to find the optimal mixed strategy for Rhona. Some lost marks for not drawing accurate graphs of expected values for $0 \leq p \leq 1$. Although it is acceptable to use the horizontal lines printed in the answer booklet, at least one of these vertical lines should have a clear scale so that the accuracy of the process of finding the highest point in the region can be checked by the examiner.

In (b)(ii) those finding the correct optimal strategy in part (b)(i) were usually able to find the correct value of the game.

Q6.

- (a) Explanations rarely scored full marks. In order to show that the game has a stable solution, it was expected that the maximum values in the columns would be indicated before finding the minimum value of these maxima. The maximum of the row minima also needed to be found. Some statement should then have been made indicating that these two values were equal and hence that the game has a stable solution. Quite a few candidates felt that all they needed to do was to draw arrows pointing to -3 when an explicit statement was required regarding the play-safe strategies for each player.
- (b)
 - (i) Most candidates were able to find the optimal mixed strategy for Rohan. Very few made a statement such as “Let Rohan play R_1 with probability p ”; usually the letter p was introduced with no explanation. Some lost marks for not drawing accurate graphs of expected values for $0 \leq p \leq 1$. Although it is acceptable to use the horizontal lines printed in the answer book, at least one of these vertical lines should have a clear scale so that the accuracy of the process of finding the highest point in the region can be checked by the examiner. Those finding the correct optimal strategy were usually able to find the correct value of the game, but for full marks it was necessary to make a statement regarding the frequencies of playing the two strategies.
 - (ii) A large number of candidates were not prepared for this part of the question. Despite the wording in the question, very few made the correct start using the

three probabilities p , q and $1 - p - q$ in order to obtain a pair of simultaneous equations in p and q . Those who made a correct start usually completed and found the optimal mixed strategy for Carla; some candidates, however, did obtain values for p and q which were either negative or more than 1.

Q7.

Some candidates argued correctly, in part (a), that row 2 dominated row 3 but then incorrectly that one column dominated another in an attempt to reduce the matrix to a 2 by 2. In order to show that the game had a stable solution, it was expected that the minimum values in the rows and maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values were equal and hence that the game has a stable solution. The actual play-safe strategies for Ann and Bill were sometimes stated incorrectly even when the correct minimax and maximin values had been identified in the table.

The method of finding the optimal mixed strategy in part (b) seemed to have been clearly understood by most candidates. Sometimes the graphs were inaccurate and a mark was lost.

Other good solutions were sometimes spoiled by not stating how often Russ should play each of the strategies R_1 and R_2 . Those who found the correct probabilities in part (b)(i) were usually able to find the value of the game in part (b)(ii), but some substituted into the wrong expression for expected gain.

Q8.

In part (a), in order to show that the game had a stable solution, it was expected that the minimum values in the rows and maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values were equal and hence the game has a stable solution.

Poor algebraic skills in part (b), together with the omission of brackets, caused many candidates to write down expressions which were often simplified incorrectly. Many failed to make progress because they failed to equate each of the expressions found in part (b)(i) to $8/3$; others obtained nonsensical answers such as $p = 4/3$ because of earlier errors in their equations. It was surprising how many took more than a page of working to solve these two simple linear equations.

Q9.

In part (a), few candidates mentioned that, **for each outcome**, one player's gain plus the other player's gain is zero in a zero-sum game.

In part (b), most responses indicated that C_1 was Colin's play-safe strategy. However, the accompanying explanations were often poor with many producing the standard table indicating that there was no stable solution when this was not required. Better candidates gave the column maxima, namely 2, 5 and 4, together with a comment about 2 being the minimum of these column maxima.

In part (c), the idea that R_3 was dominated by R_1 was well understood, but some candidates spoiled their answer by suggesting that strategy R_3 was worse than both R_1 and R_2 .

Despite the standard nature of this question, it was surprising to see many candidates unaware of how to find the optimal mixed strategy in part (d). A good sketch showing the feasible region was expected with the highest point of the feasible region being selected in order to find the probability of playing the various rows. Many gave no reason for introducing a probability p nor explained what the actual mixed strategy was when the value of p had been found. Many candidates calculated the value of the game, which was not required

Q10.

In part (a), in order to show that the game has a stable solution, it was expected that the minimum values in the rows and maximum values in the columns would be indicated before finding the maximum of the minima and the minimum of the maxima. Some statement should then have been made indicating that these two values are equal and hence the game has a stable solution. To find the optimal mixed strategy in part (b), when a letter such as p is introduced, there should be an indication that this is the probability that Ros is choosing R_1 , for example. Three expressions in p were often written down with no indication as to what they represented. Although three linear graphs were often drawn, many candidates chose the wrong pair to solve for the optimal strategy and seemed to be guessing rather than reasoning correctly which point to select. Those who obtained the correct optimal strategy were usually able to find the value of the game.

Q11.

Part (a)(i) proved to be the most testing part of this question, with very few candidates giving a good reason why Rose should choose R_1 as the play-safe strategy. The minimum value in each row had to be calculated and then the maximum of these minima needed to be shown to equal -1 . In part (a)(ii), the better solutions gave a table with the column maxima 5, 2 and 5 clearly stated and with the minimum of these values indicated. A statement that the minimax value, 2, was not equal to the maximin value, -1 , from part (a)(i) was then required to show that the game has no stable solution.

In part (b), most candidates realised that R_1 dominates R_3 and showed that corresponding entries of R_1 were greater than those of R_3 .

The expressions for the expected gains in part (c) were usually correct, although some made algebraic slips when trying to simplify their expressions. There was a casual approach by many when drawing the graphs. Lines were often too short or too long and many made no attempt to indicate values on the vertical axis. Candidates do not appear to know what the three lines on the graph represent. Some seemed to guess which two expressions to equate when finding the optimal mixed strategy for Rose, instead of considering the highest point in the feasible region.

It was necessary to make a statement indicating that Rose plays R_1 for $2/3$ of the time and R_2 for $1/3$ of the time in order to score full marks in part (c)(iii). It was quite surprising to see a number of candidates substituting $p = 2/3$ into their third (unused) expression and consequently obtaining the wrong value of the game.

Q12.

The idea of dominance seems to be well understood. Marks were lost in part (b) for not identifying the actual play safe strategies. It is not sufficient to draw circles around a couple of numbers and to simply state there is no saddle point. Unfortunately, this is precisely what many candidates offered as their solution. This question was a simple example of game theory, reducing to a 2 by 2 game, and most candidates wrote down and solved correctly an equation in p and hence the expected points gain for Sam.

Consequently, it was pleasing to see many high marks in this question.

Q13.

Candidates knew the theory for part (a) of this question but there were very few scripts that earned full marks. Candidates had to show calculations to answer this question rather than simply stating that there is no play safe solution.

Part (b) was well answered.

Part (c) was the least well answered question on the paper. Some candidates adopted the correct approach, using probabilities p , q , and $1 - p - q$, but still found the completion of the question difficult. There were, however, a number of correct solutions.



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.